UNIT 5
APPLICATION OF COMPUTER SCIENCE

11TH COMPUTER SCIENCE FEDERAL BOARD

AWINASH GOSWAMI

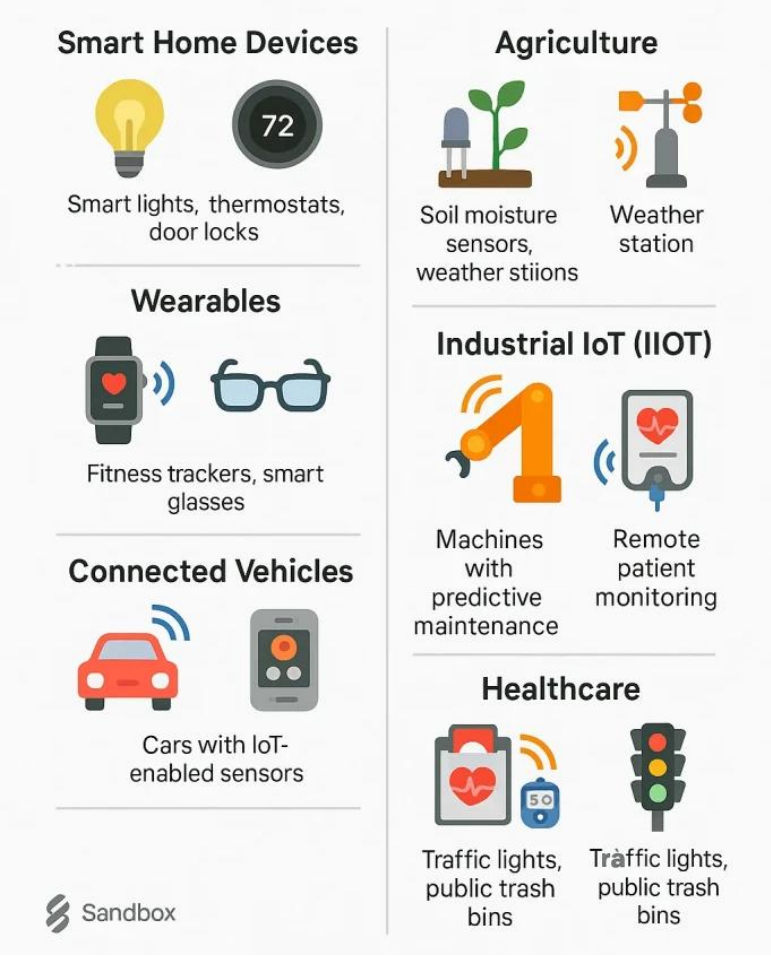
AGENDA

- Introduction to IoT

INTRODUCTION TO IOT

- IoT means everyday objects connected to the internet so they can send and receive data.
- Examples:
  - A smartwatch that tracks your steps and sends them to your phone.
  - A fridge that tells you when you're out of milk.
  - A doorbell camera you can check from your phone
  - A thermostat you control with an app.

So basically, "things" (not just computers or phones) that can talk to the internet and to each other.

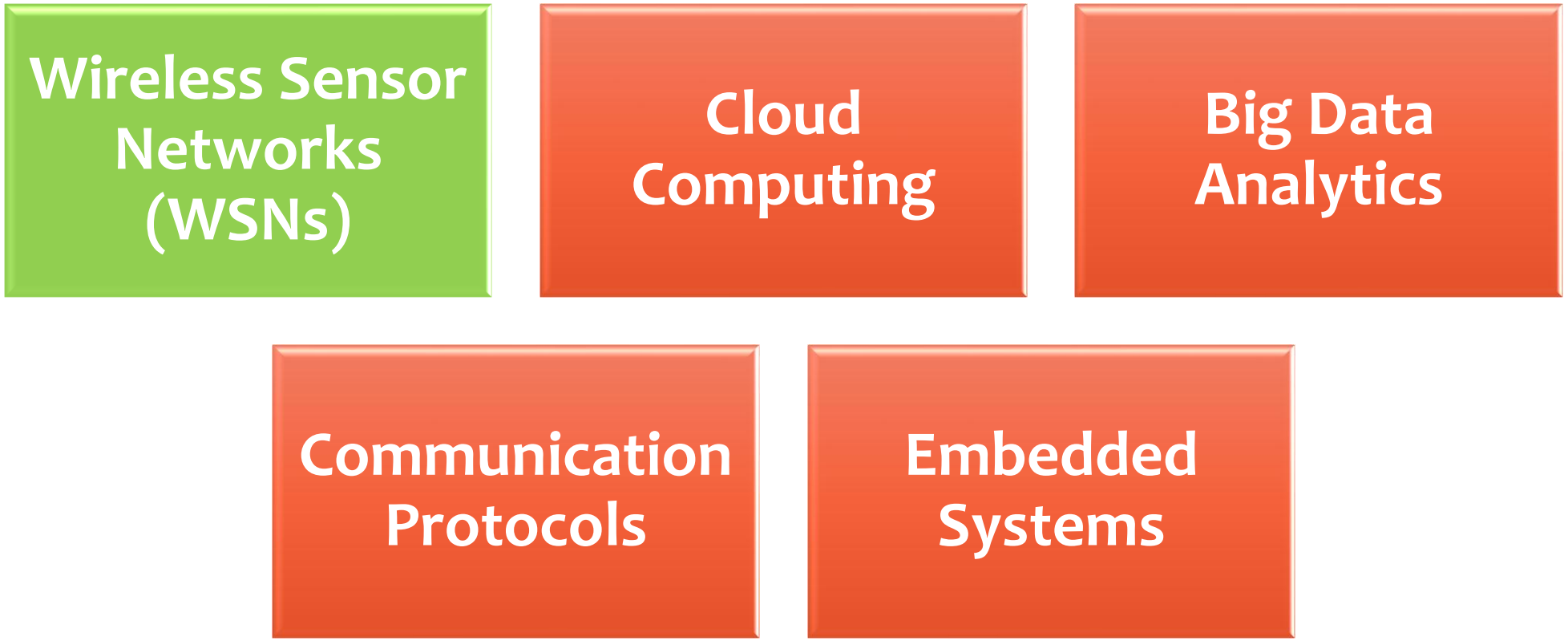


INTRODUCTION TO IOT CONT...

- **Technically, IoT** is a network of physical devices (often run on microcontrollers with limited CPU, memory, and power) embedded with **sensors and connectivity hardware** that collect and exchange data over network protocols — typically without requiring human intervention.

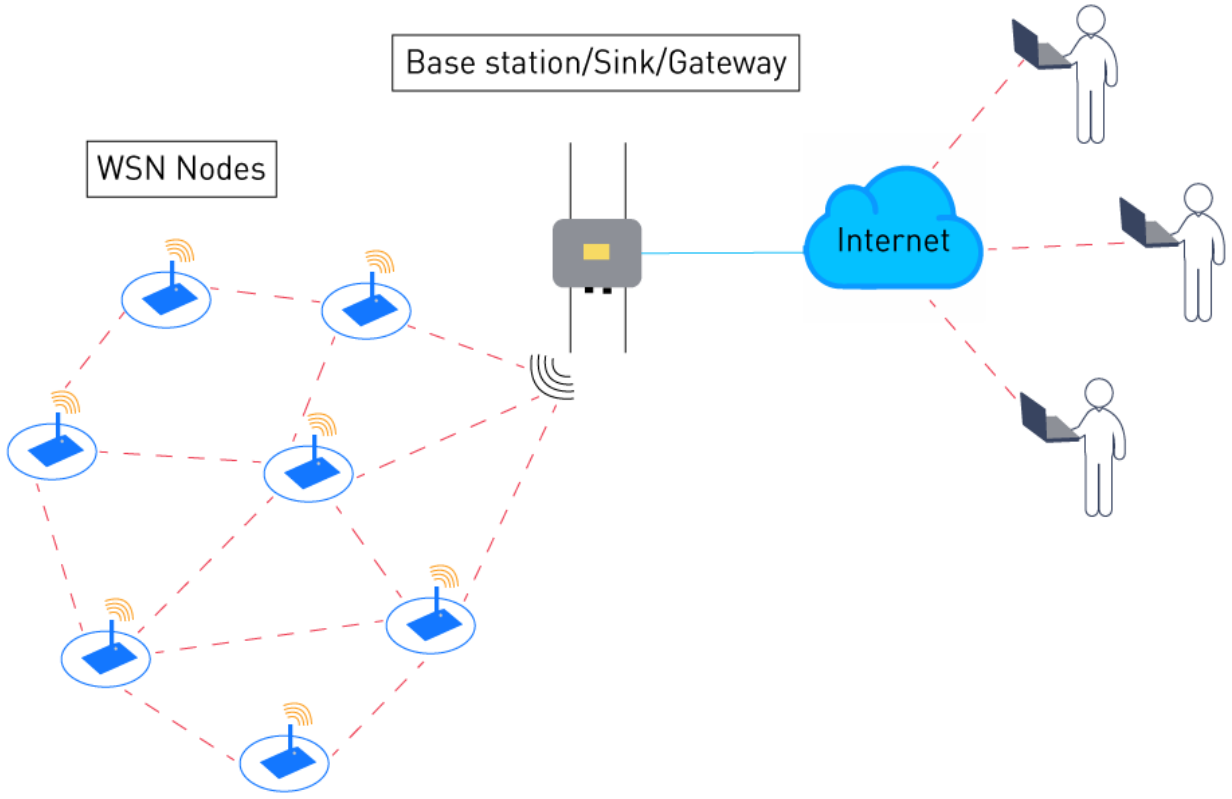
- **Core Architecture (typically 4–5 layers)**
  - **Perception/Device Layer** — Sensors (temperature, motion, GPS, etc.) that interact with world.
  - **Network/Connectivity Layer** —Bluetooth, Wi-Fi
  - **Cloud/Platform Layer** — AWS IoT, Azure IoT Hub for device management and storage.
  - **Application Layer** — Dashboards, APIs (smart home, industrial IoT, healthcare).

TECHNOLOGIES THAT ENABLED INTERNET OF THINGS



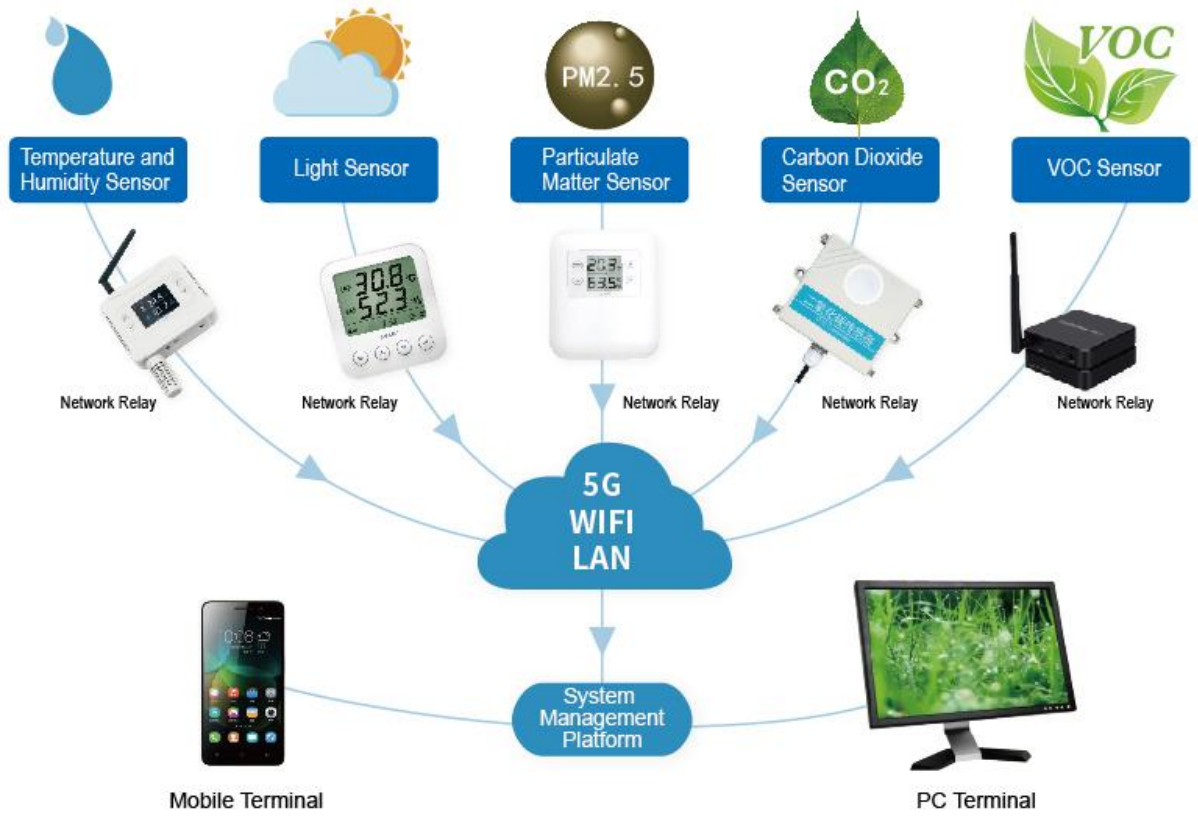
WIRELESS SENSOR NETWORKS

- It is a network of IoT physical devices that communicate through wireless channels.
- The devices of a WSN are embedded with sensors and software.
- Three types of sensors are used to monitor environmental and physical conditions.
- These are **environmental sensors, industrial sensors** and **motion detection sensors.**



WIRELESS SENSOR NETWORKS: **ENVIRONMENTAL SENSORS**

Environmental sensors are used for measuring and monitoring environmental conditions i.e. temperature, humidity, air quality, air pressure, wind speed and light sensors.



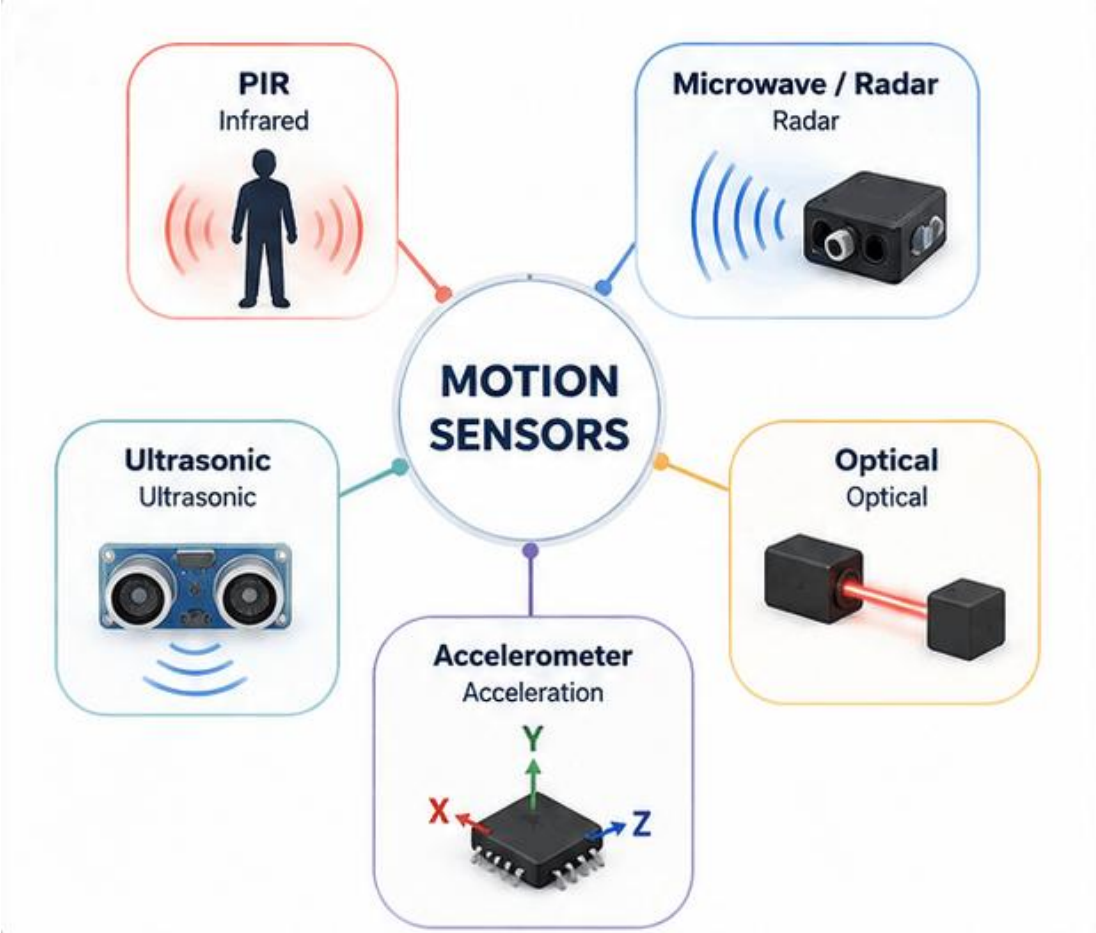
WIRELESS SENSOR NETWORKS: **INDUSTRIAL SENSORS**

Industrial sensors are used for measuring and monitoring physical qualities in manufacturing process. The physical qualities include heat, pressure, gas, vibration, viscosity, fluid flow, and magnetitic field sensors.



WIRELESS SENSOR NETWORKS: **MOTION DETECTION SENSORS.**

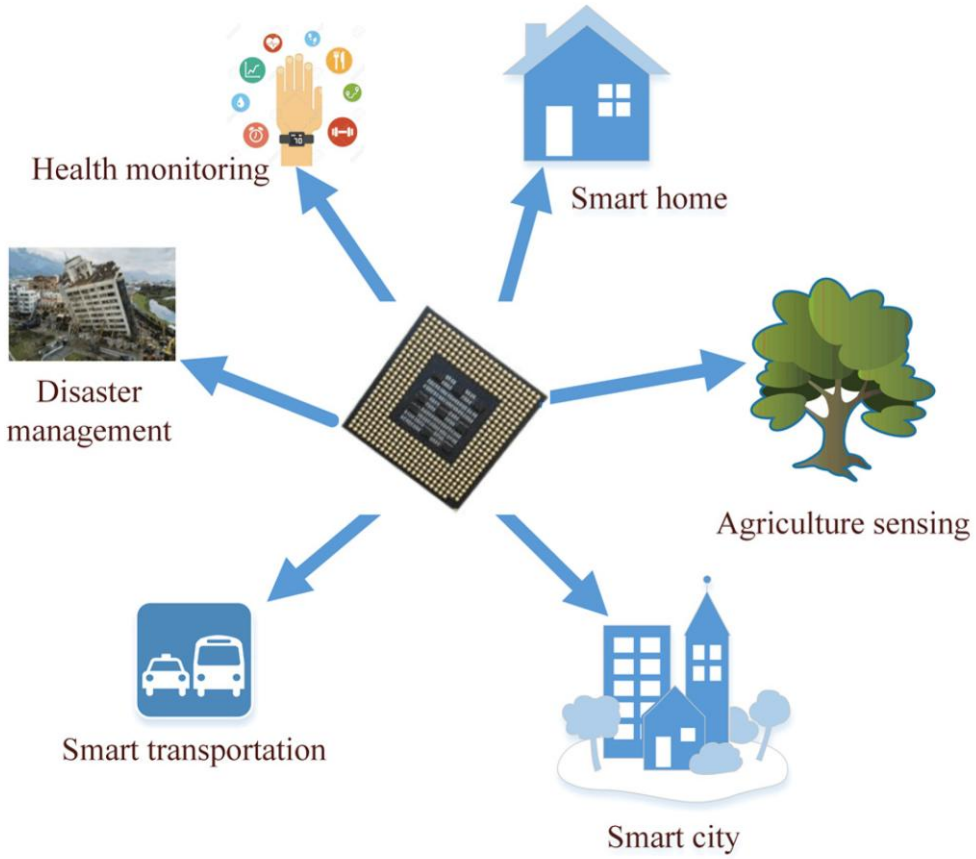
Motion detection sensors are used to detect movement of human or objects in its surrounding. These are commonly used to prevent unauthorized movement in restricted areas. The motion detection sensors include ultrasonic, proximity, vibration, Passive Infrared sensors.



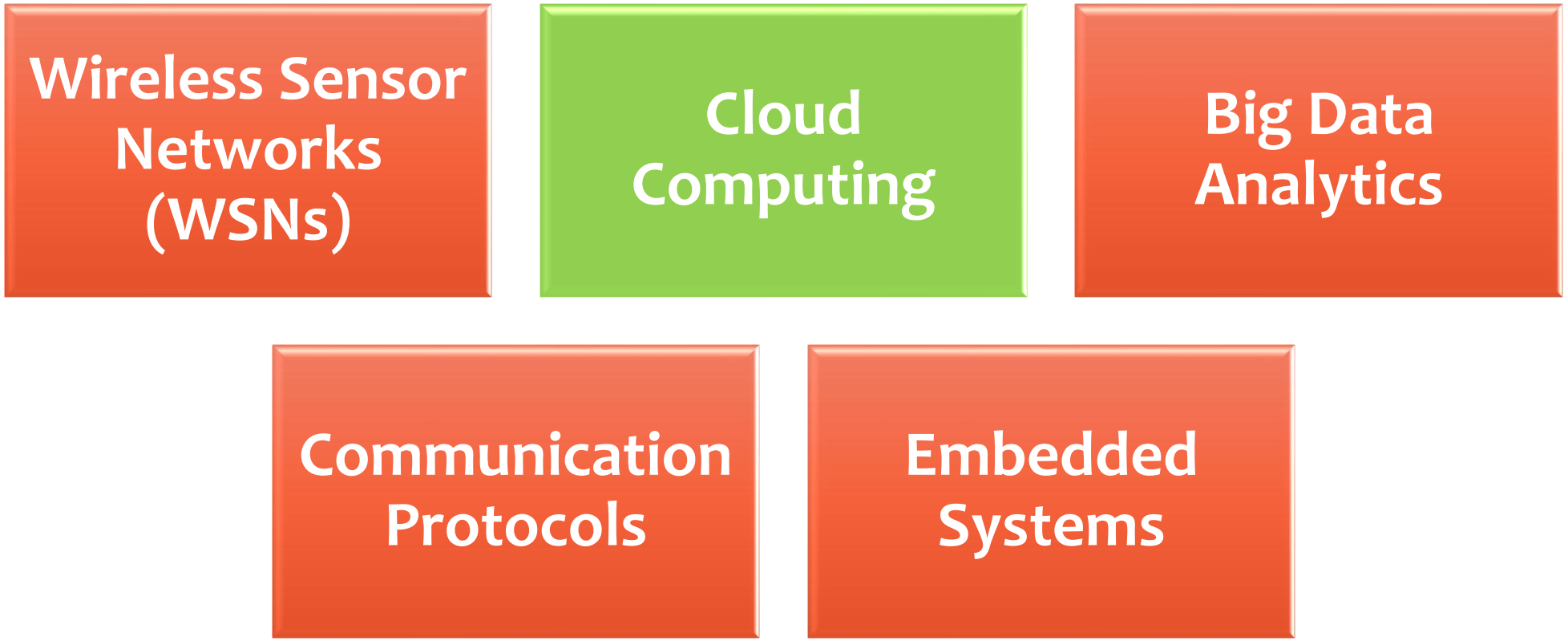
WIRELESS SENSOR NETWORKS CONT.

The application of WSN IoT systems are:

- Surveillance System
- Weather Monitoring System
- Health Monitoring System
- Smart Grid Electrical Network



TECHNOLOGIES THAT ENABLED INTERNET OF THINGS



CLOUD COMPUTING

It is a service to provide computer resources (storage, database, server, processor, RAM etc) remotely via internet.

There are four types of cloud computing models;

- Public cloud
- Private cloud
- Community cloud
- Hybrid cloud



CLOUD COMPUTING: PUBLIC CLOUD

- **Public cloud** is computing power and storage that a company rents from someone else, delivered over the internet, and shared with other customers.
- Analogy: you rent an apartment in a big building. The building (servers) is owned by someone else (like Amazon, Google, or Microsoft), you share the infrastructure with other tenants, but your apartment (your data and apps) is private to you.
- **Key points:**
  - **Owned by a provider** — Amazon Web Services (AWS), Microsoft Azure, Google Cloud, etc.
  - **Shared infrastructure** — multiple customers use the same physical servers, but their data is isolated from each other.
  - **Pay-as-you-go** — you pay only for what you use, no upfront hardware costs.
  - **Accessible anywhere** — you reach it over the internet from any device.
  - **Scales easily** — need more power? Add it in minutes instead of buying servers.

CLOUD COMPUTING: PRIVATE CLOUD

- **Private cloud** is computing power and storage that is dedicated to a single organization — not shared with anyone else.
- Analogy: you own or exclusively rent the whole building. Only your organization uses it.
- **Key points:**
  - **Single tenant** — only your organization uses the infrastructure; no other customers share it.
  - **More control** — you control security, configuration, and compliance more tightly.
  - **More expensive** — you pay for the hardware/infrastructure even when you're not using it fully.
  - **Less elastic** — scaling up takes longer than in a public cloud, since you're limited by your own hardware.
  - **Simple example:** A bank runs its own data center with virtualization software (like VMware) so its teams can spin up servers on demand — but only the bank's employees use it, and the data never leaves the bank's control.
- **Why choose it?** Strict regulations, sensitive data, or a need for full control.
- **Why avoid it?** Higher cost and less flexibility than public cloud.

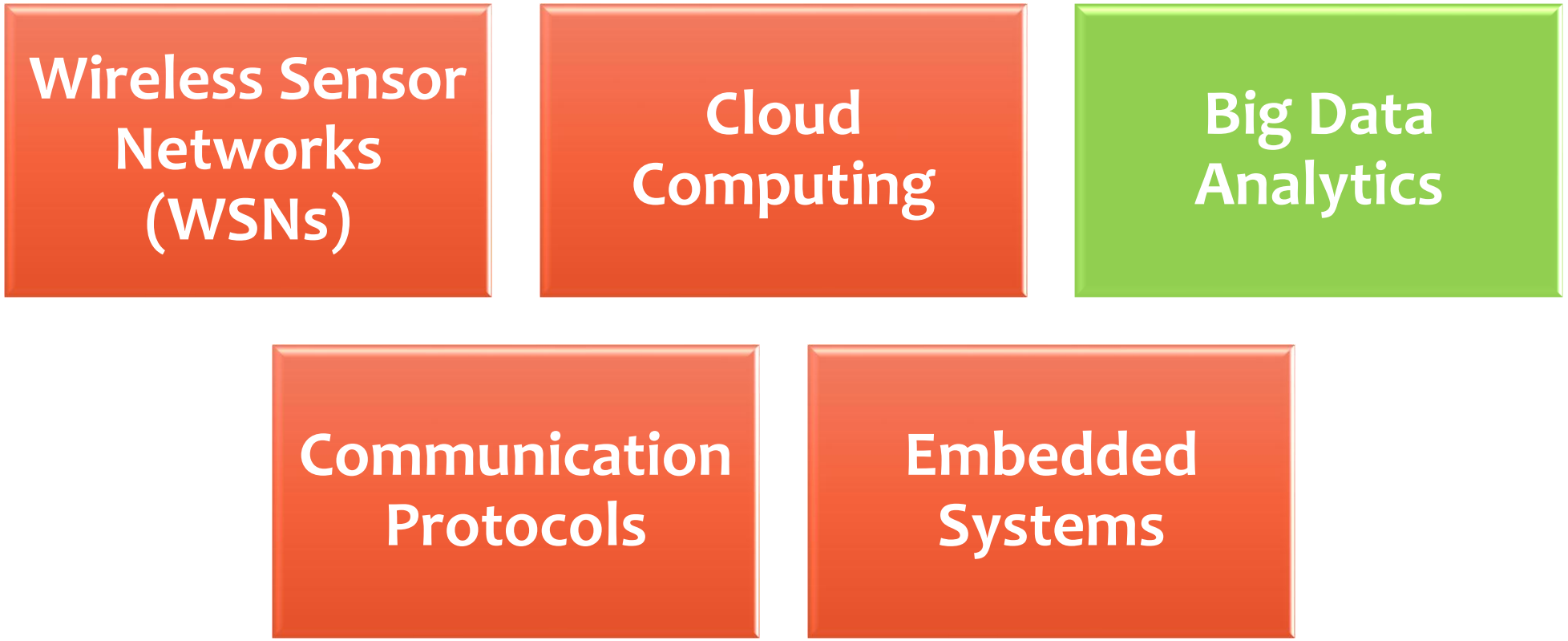
CLOUD COMPUTING: COMMUNITY CLOUD

- **Community cloud** is computing infrastructure that is shared by a specific group of organizations that have similar needs — but it's not open to the general public.
- Analogy: a gated community built for a specific group (like a retirement village or a university campus). The residents share the facilities because they have common needs, but outsiders can't just move in.
- **Key points:**
  - **Shared by a specific group** — multiple organizations use the same infrastructure, but only those in the "community" (e.g., banks, government agencies, healthcare providers, or universities).
  - **Common concerns** — the group shares needs like strict security, compliance rules, or mission requirements (e.g., all must follow the same banking regulations).
  - **Owned and managed** — it can be run by one of the organizations in the group, all of them together, or a third-party provider.
  - **On-site or off-site** — it can live in a shared data center or be hosted elsewhere.
  - **Cost-effective** — because the cost is split among several organizations, it's cheaper than each building its own private cloud.
  - **Security middle ground** — more secure than a public cloud (because access is restricted to the community), but less isolated than a private cloud (because you still share with others).
- **Simple example:** Several hospitals create a shared cloud to store patient records. They all must follow the same health-data privacy laws, they share the cost of the infrastructure, and only those hospitals can access it — no random company can join.

CLOUD COMPUTING: HYBRID CLOUD

- **Hybrid cloud** is a setup that combines a public cloud and a private cloud, connected together so they can share data and applications.
- Analogy: a **house with an attached garage or a home office**:
- **Key points:**
  - **The "Best of Both Worlds"** — you get the tight security and control of a private cloud, plus the cheap, massive scale and flexibility of a public cloud.
  - **Data mobility** — technology connects the two environments, allowing data and applications to move between them smoothly.
  - **The "Split" strategy** — organizations divide their work. They put highly sensitive data (like customer financial records or trade secrets) on the **private cloud**, and put less sensitive, public-facing tasks (like a company website or testing new apps) on the **public cloud**.
  - **Cost and speed** — you save money by using the public cloud for heavy, temporary workloads, but you get fast, secure access to your critical data on the private side.
- **Simple example:** An online retailer. During normal months, their website runs on their **private cloud**. But on Black Friday, traffic explodes. They use a **hybrid cloud** setup to automatically burst into the **public cloud** to handle the extra shoppers, while keeping all customer credit card data safely locked away in their **private cloud**.

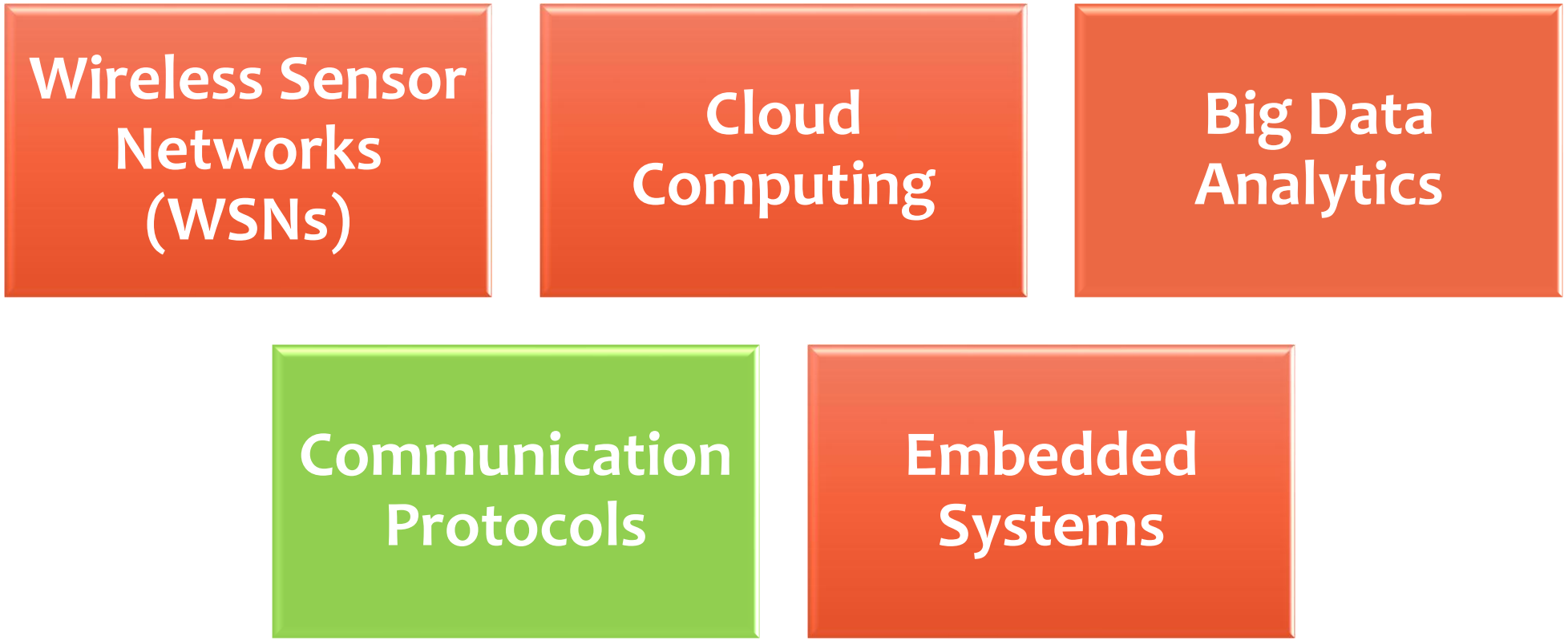
TECHNOLOGIES THAT ENABLED INTERNET OF THINGS



BIG DATA ANALYTICS

- Big data analytics is concerned with collecting extremely big data from various resources and using analytics software to find out market trends, patterns and insights.
- Huge volume of data is created in our daily life from various digital resources. These include banking transactions, customer databases, medical records, emails, social media apps, etc. which cannot be handled by traditional databases.
- The data gathered by IoT devices is transformed into actionable information by big data analytics to improve decision-making.
- Some areas of big data analytics are; Healthcare industry, Transport industry, Banking and financial services, Weather patterns and trends, Insurance industry, Digital marketing, Media and entertainment, Business insights, Government sector.

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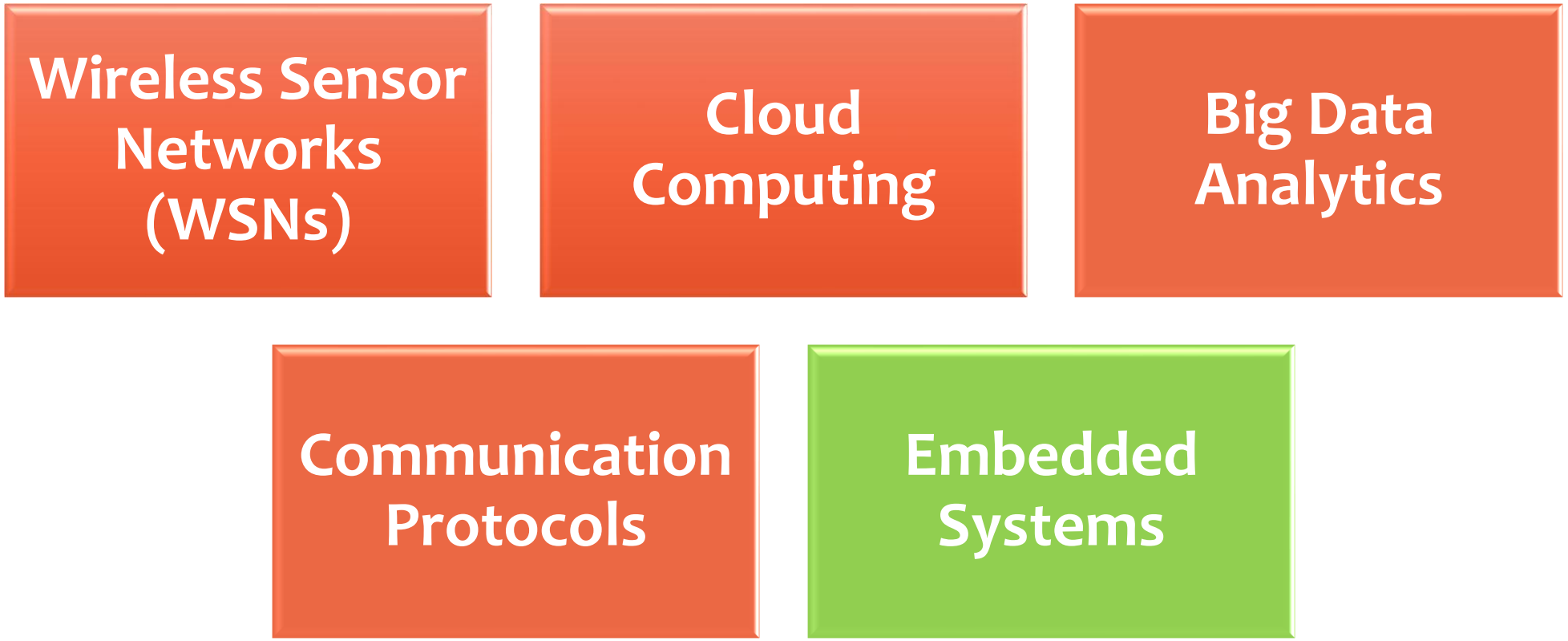


COMMUNICATION PROTOCOLS

- IoT communication protocol is a set of rules that govern how data is transferred and understood between physical devices, sensors and the end users.

- The common examples of protocols are Wi-Fi, Bluetooth, HTTPS, Zigbee etc.

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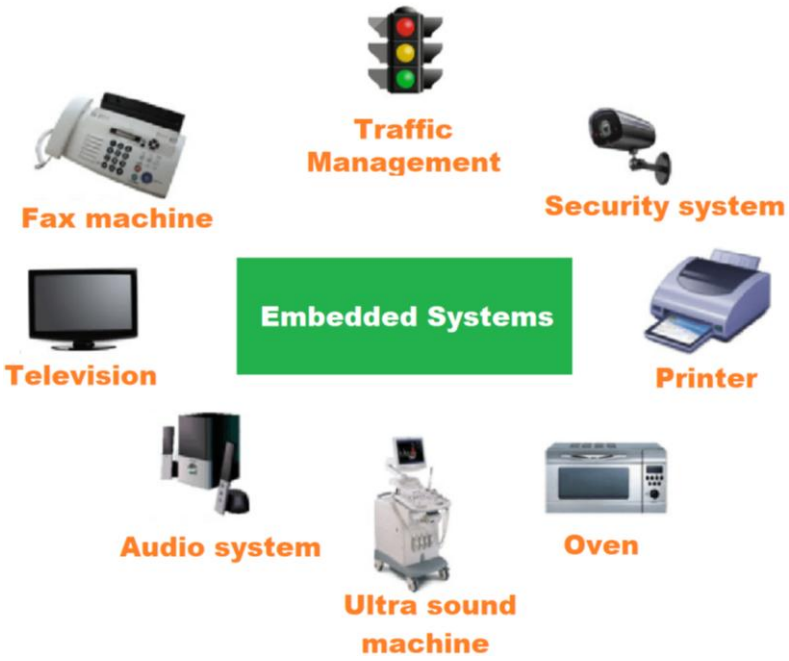
EMBEDDED SYSTEMS

- Embedded systems are small electronic devices which are combination of hardware and firmware (*a software that is permanently installed in a device*) designed to perform a specific task.

- Embedded systems have a microcontroller or a microprocessor that controls the operation of embedded systems through firmware.

- Embedded systems have various applications in IoT such as they can monitor a room's temperature and automatically adjust the thermostat.

- **Embedded Systems Examples:** Traffic Management, Medical Equipment, Ticket Scanner, Vending Machine, Fax machine, Security system, Television, Printer, Audio system, Ultra sound machine, Oven



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